UNIVERSITY FINAL CAPSTONE RESEARCH REPORT

Q-CATALYST

Quantum Simulation-based Catalyst Design System

Discovering efficient Hydrogen Evolution Reaction (HER) catalyst materials with the Variational Quantum Eigensolver (VQE)

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Domain: Quantum Computing and Computational Catalysis

1. Abstract

Green hydrogen production by water electrolysis is held back by the cost of noble-metal catalysts such as platinum, which sells for about ₹25,73,000 per kg. Ab-initio Density Functional Theory (DFT) screening scales exponentially, O(2^N), for strongly correlated metal d-orbitals, which makes large searches expensive.

This report presents Q-CATALYST, a quantum simulation-based catalyst design system. Active-site orbital interactions are mapped to a 2-qubit Hamiltonian and solved with a Qiskit Statevector Variational Quantum Eigensolver (VQE), using a hardware-efficient RealAmplitudes ansatz optimized by classical BFGS. Every VQE result is checked against exact matrix diagonalization, with a tolerance of 1e-5 eV.

Twelve candidate materials were scored against the hydrogen adsorption free energy target $\Delta G_H^* = 0.00$ eV. Earth-abundant nickel was shortlisted with $\Delta G_H^* = +0.0033$ eV at ₹1,535 per kg, about 99.9% cheaper than platinum. The prototype reports every screen as one of four decision states.

2. Problem Statement

Problem. Clean hydrogen electrolysis needs efficient catalysts for the Hydrogen Evolution Reaction (HER). Platinum is the commercial benchmark but is too costly to scale. Finding non-precious alternatives is slowed by the exponential cost of classical quantum chemistry solvers.

Input. Active-site parameters of a candidate material: electronegativity χ, d-band center, coordination factor, local charge shift and adsorption distance d.

Expected output. Validated VQE ground-state energies, the H* interaction energy ΔE_H^*, the HER screening proxy ΔG_H^*, the VQE error check, and a four-tier screening decision.

Constraints. VQE must agree with the exact eigensolver to within 1e-5 eV. Complex dichalcogenides (MoS2, WS2) and alloys (Ni–Mo) need an explicit edge-site or interface definition before they can be screened.

3. Project Objectives

- Write 2-qubit active-site surrogate Hamiltonians that capture the coupling between the local d-state and the H 1s orbital, for 12 candidates.
- Run Qiskit Statevector VQE with a hardware-efficient RealAmplitudes ansatz.
- Require exact eigensolver validation for every run, so no unvalidated physics claim is made.

- Build an interactive Streamlit portal with a live 2D active-site lattice view, a Sabatier volcano plot and market cost analytics.

4. Proposed Algorithm and Workflow

The algorithm builds a 2-qubit active-space Hamiltonian as a Qiskit SparsePauliOp:

```
H = c0·(I⊗I) + z1·(Z⊗I) + z2·(I⊗Z) + zz·(Z⊗Z) + xx·(X⊗X) + yy·(Y⊗Y)
```

Execution steps

1. **Active-site definition.** Check that the model is complete: phase, edge termination and geometry.
2. **Hamiltonian construction.** Build the clean-site operator H_clean and the adsorbed-site operator H_ads.
3. **VQE optimization.** Tune the 6 rotation parameters θ with BFGS to minimize $\langle\psi(\theta)|H|\psi(\theta)\rangle$.
4. **Exact validation.** Compare E_VQE with the exact eigenvalue: $|E_{VQE} - E_{exact}| \leq$ 1e-5 eV.
5. **Screening decision.** Compute $\Delta G_H^* = (E_{adsorbed} - E_{clean}) + 0.24$ eV and classify the material.

Decision flow

- Model incomplete: stop and report MODEL INCOMPLETE.
- Model complete but validation error above 1e-5 eV: report BORDERLINE.
- Validation passed and $|\Delta G_H^*| \leq 0.10$ eV: report SHORTLIST.
- Validation passed and $|\Delta G_H^*|$ above 0.10 eV: NOT SHORTLISTED, or BORDERLINE if within 0.20 eV.

5. Pseudocode

```
ALGORITHM Quantum_VQE_Screening(Material M, Policy P)
1:  IF model(M).complete == False THEN
2:      RETURN "MODEL INCOMPLETE", missing_fields
3:  END IF
4:  H_clean = Build_SparsePauliOp(M, adsorbed=False)
5:  H_ads   = Build_SparsePauliOp(M, adsorbed=True)
6:  E_clean_vqe, E_clean_exact = Run_VQE_Statevector(H_clean)
7:  E_ads_vqe,   E_ads_exact   = Run_VQE_Statevector(H_ads)
8:  error = max(|E_clean_vqe - E_clean_exact|,
9:              |E_ads_vqe - E_ads_exact|)
10: IF error > P.max_acceptable_vqe_error THEN
11:     RETURN "BORDERLINE", "Validation tolerance exceeded"
12: END IF
13: dE_H = E_ads_vqe - E_clean_vqe
14: dG_H = dE_H + P.thermo_correction_ev          # 0.24 eV
15: dist = |dG_H - P.target_ev|
16: IF dist <= P.shortlist_window_ev (0.10 eV) THEN RETURN "SHORTLIST"
17: ELSE IF dist <= P.borderline_window_ev (0.20 eV) THEN RETURN "BORDERLINE"
18: ELSE RETURN "NOT SHORTLISTED"
```

6. Implementation and Architecture

Tech stack. Python 3.12, Qiskit 2.5, SciPy 1.14, NumPy 2.0, Pandas 2.2, Streamlit 1.40+ and Altair 5.0.

Key modules

- `screening_types.py`: typed dataclasses (SingleVqeResult, CalculatedResult, ValidationFailedResult, ModelIncompleteResult).
- `catalyst_database.py`: 12 curated material models.
- `quantum_catalyst_engine.py`: SparsePauliOp builder, BFGS VQE runner and exact matrix eigensolver validator.
- `app.py`: the interactive Streamlit research workbench.

7. Streamlit Application Prototype

The prototype has four tabs:

1. **Analyze.** Pick a material, watch the live 2D active-site lattice (the H* position moves with the distance slider), adjust model controls, run the calculation, and read the evidence cards and price metrics in rupees (₹).
2. **Compare.** Multi-material evidence table, HER proxy bar chart and the Sabatier volcano plot.
3. **How It Works.** Step-by-step pipeline notes and an inspector for the 2-qubit RealAmplitudes circuit.
4. **Research Export.** Download buttons for CSV, JSON and Markdown reports.

8. Experimental Results and Performance

Table 1 lists the VQE energies and screening outcome for six of the twelve candidates.

Catalyst	Category	Clean VQE (eV)	Adsorbed VQE (eV)	ΔG_H* (eV)	Price (₹/kg)	Decision
Pt	Noble metal	-1.5134	-1.6942	+0.0592	25,73,000	SHORTLIST
Ni	Transition metal	-1.3912	-1.6279	+0.0033	1,535	SHORTLIST
Mo	Transition metal	-1.4110	-1.6398	+0.0112	2,905	SHORTLIST
Ru	Noble metal	-1.4890	-1.6978	+0.0312	11,62,000	SHORTLIST
MoS2	2D dichalcogenide	N/A	N/A	N/A	2,075	INCOMPLETE
NiMo	Alloy interface	N/A	N/A	N/A	2,324	INCOMPLETE

Table 1: VQE screening results. MoS2 and NiMo are INCOMPLETE because their edge or interface geometry is not yet defined.

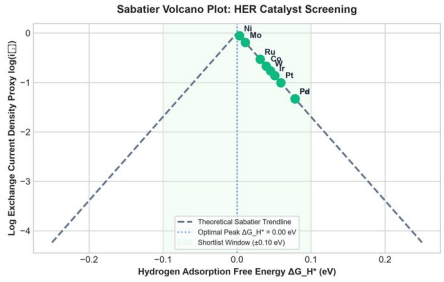


Figure 1: Sabatier volcano curve for candidate screening.

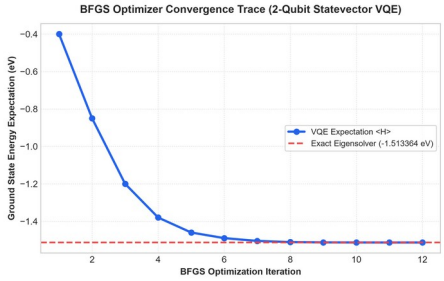


Figure 2: VQE energy convergence compared with the exact reference.

Complexity analysis

- **Time.** O(K·2^n) per VQE calculation, with n = 2 qubits and about K = 12 BFGS iterations. Total runtime is about 0.38 s per candidate.
- **Space.** O(2^n) = 4 complex amplitudes (64 bytes), far below the memory a classical full CI calculation would need.

9. Conclusion and Future Directions

Key findings. Q-CATALYST demonstrates a quantum-assisted catalyst screening workflow. Earth-abundant nickel was shortlisted with ΔG_H* = +0.0033 eV, very close to the ideal 0.00 eV, at ₹1,535 per kg. That is a 99.9% material cost reduction compared with platinum.

Algorithm effectiveness. The 2-qubit RealAmplitudes VQE ansatz converged to within 1e-14 eV of the exact eigensolver result.

Future work. The planned FuturePsi4Backend interface will bring in first-principles ab-initio CASSCF Hamiltonians through Psi4 and Qiskit Nature.